Aditi Gode

812-802-3760 | adigode@iu.edu | linkedin.com/in/aditi-gode | github.com/aditigode

EDUCATION

Master of Science in Computer Science, Indiana University Bloomington — Aug. 2021 – May 2023

***Courses**: Applied Machine Learning, Computer Vision, Cloud Computing, Artificial Intelligence, Database Design, Security for Networked Systems, Advanced Operating Systems, High Performance Computing;* ***GPA:** 3.9/4*

Bachelor of Engineering in Computer Engineering, University of Mumbai — Jul. 2014 – May 2018

***Courses**: Data Warehousing and Mining, Software Engineering, Object Oriented Programming, Artificial Intelligence, Machine Learning;*
GPA: 8.93/10, **Department rank: 1**

TECHNICAL SKILLS

Languages: Python, C, SQL, Java, PowerShell scripting
Distributed computing: AWS, Hadoop streaming, PySpark, Docker
Web Development: HTML, CSS, JavaScript, Beautiful Soup, PHP, MongoDB, Flask, Git, RESTful API
Machine Learning: TensorFlow, Keras, NLTK, Scikit-Learn, Spacy, Pandas, NumPy, Matplotlib, Seaborn, SciPy

PROFESSIONAL EXPERIENCE

Teaching Assistant, *Indiana University Bloomington*, *Indiana* — Aug 2022 – Dec. 2022

- **Mentored 140+ students** in a graduate-level **Applied Machine Learning** course, providing guidance during office hours and on dedicated Q&A platforms, while ensuring accurate evaluation of their submissions
- Technologies used: Scikit-Learn, Pandas, NumPy, TensorFlow

Research Assistant, *Indiana University Bloomington*, *Indiana* — May 2022 – Jul. 2022

- Developed a data browsing and data visualization tool using **Streamlit.io** to study Blink dynamics
- Performed **peak shape analysis** of segmented blinks to prepare the data for classification using Machine Learning tools
- Technologies used: Streamlit.io, Python, NumPy, Pandas, SciPy

System Engineer, *Citibank(contract)*, *Mumbai, India* — Nov. 2019 – Apr. 2021

- Demonstrated exceptional proficiency in **Windows server administration, network configuration, and Linux server administration**, placing within the **top 3%** of training candidates
- Developed automation scripts using PowerShell to streamline Citibank's filesystem checks, achieving an **85% reduction in missing data incidents**
- Collaborated with multiple tech teams to resolve an average of **11 high-priority server storage and security incidents** daily, leveraging ServiceNow for incident management
- Harnessed MS Excel to enhance data insights, driving a **20% boost** in strategic decisions through trend analysis
- Technologies used: Windows servers, PowerShell, ServiceNow, Microsoft Excel

System Engineer, *StateStreet(contract)*, *Bangalore, India* — Feb. 2019 – Nov. 2019

- Spearheaded the management of **StateStreet's Enterprise Data Warehouse (EDW)**, consistently maintaining a **98% success rate** in meeting SLA-driven client data extraction timelines
- Optimized **SQL for data querying** to **identify and rectify instances of missing data**, enhancing data integrity and overall system stability
- Technologies used: Oracle SQL Developer, Linux, Autosys

PROJECTS

ML inferences on data using serverless functions | *AWS Lambda, Rekognition, S3, CloudWatch, ECR, Translate, TensorFlow, Docker, Flask, Boto3*

- Implemented serverless for ML inference using AWS Lambda, **optimizing latency with a Multi-Buffer System (MBS) simulation** for NLP tasks
- Developed a user-friendly UI interface for seamless object detection, text translation, and concurrent file management using Flask and Boto3
- Achieved **15% cost-saving and 30% reduction in inference time** by distributing tasks across varied Lambda memory configurations

Distributed Analysis of large datasets with Spark | *PySpark, Data Visualization, Spark-SQL, HDFS*

- Utilized PySpark on Indiana University's Jetstream2 running OpenStack to effectively analyze **1.8 GB** of NYC parking violation data, identifying key ticketing patterns and peak issuance days
- Leveraged **K-means clustering in PySpark** for NBA player shooting pattern analysis, achieving a **silhouette score of 0.703** for optimal cluster quality

Emotion Recognition in Toddlers for Social Robot Interaction | *TensorFlow, Keras, Python, NumPy, FER2013 dataset, Google Colab*

- Engineered and optimized multiple neural network architectures, achieving a **79.71% training accuracy** with **VGG16**, and a **testing accuracy of 68%**, outperforming other tested models like **ResNet18** and **Convolutional Compact Transformer**
- Devised strategies to address dataset imbalance and overfitting, notably enhancing performance by introducing **batch normalization, which expedited training significantly, and using dropout layers to combat overfitting**

Automatic prediction of tags using NLP | *Python, Scikit-learn, Pandas, NumPy, NLTK, Exploratory data analysis*

- Solved the **multilabel classification problem** of predicting 1-3 tags for **200k StackOverflow Q&A** by extensively analyzing, and transforming the data
- Trained and evaluated the performance of models for multi-label classification using **One vs Rest classifier, Multi-label KNN, Multinomial Naïve Bayes, and Binary Relevance classifier**

Delivery Service Database | *MySQL, ERDPlus*

- Designed a database for online shopping application for buying items from local stores using **MySQL** and tested the database by querying for business-related information retrieval